

Power in Pixels, Words, and Sound: Multimodal Computational Methods in Political Science

OCR, CNN fine-tuning and Vision Transformers

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Updates on Last Session

Optical Character Recognition

CNN fine-tuning

Transformers

Vision Transformers (ViT)

Updates on Last Session: Hashing & Data Augmentation

RandomCrop / RandomFlip

- These transforms **do not create duplicates** — they crop/flip images *in place* within the dataset
- You can set a **probability** for how often an image is flipped or cropped

What is a Hash?

- A hash converts an image into a **unique digital fingerprint**
- Before hashing, we pre-process the image so that **similar (but not identical)** images can still be detected

Further Reading

- Stack Overflow: differences between hashing methods
<https://stackoverflow.com/questions/998662/what-is-image-hashing-used-for>
- Official package website: <https://www.phash.org/>

Optical Character Recognition (OCR)

OCR is the automated conversion of images containing text into machine-readable text. We will not dive into the mathematics here - this is a completely own research field with many different methods.

- **Pre-processing is essential:** Clean images (thresholding, morphological ops) dramatically improve OCR accuracy
- **OCR engines are not image processors:** → they need clean, segmented input to perform well
- **Post-processing matters:** Apply spellcheck, regex, and domain heuristics to correct inevitable OCR errors
- **Key tools:** Tesseract (most popular), pytesseract (Python wrapper), and cloud-based OCR APIs

Hyperparameters are settings chosen *before training* that control how the CNN learns - they control how fast, stable and generalizable learning is

Core training hyperparameters:

- **Epochs:** number of full passes over training data
- **Learning rate:** step size for weight updates (too high → unstable, too low → slow learning)
- **Batch size:** number of samples per update (speed vs. stability trade-off)
- **Train/validation split:** prevents overfitting via held-out evaluation

Regularization and optimization:

- **Dropout:** randomly deactivates neurons to reduce overfitting
- **Optimizer (e.g., SGD):** defines how gradients update weights
- **Momentum:** smooths updates using past gradients

Learning rate scheduling:

- **Step size:** how often learning rate is reduced
- **Gamma:** factor controlling decay strength

Does the CNN generalize well beyond the training data?

Standard Evaluation Metrics

(already covered in Part 1 of this seminar)

- **Accuracy:** overall correctness
- **Precision:** correctness of positive predictions
- **Recall:** ability to find all true positives
- **F1 score:** balance of precision and recall

Overfitting Detection

- Training loss keeps decreasing
- Validation loss stagnates or increases
→ model memorizes instead of generalizing

Model Debugging Strategies

- Inspect false positives / false negatives
- Use confusion matrix for error patterns
- Add diverse training data if bias is detected

Search Strategy

Use **Bayesian**, **Random**, or **Grid Search** to efficiently explore the hyperparameter space and avoid overfitting.

Self-Attention & Positional Encoding

- **Self-Attention:** each token generates queries, keys, and values to attend to *all* tokens simultaneously → enables parallel computation, but incurs $O(n^2)$ complexity (prohibitive for long sequences)
- Also called *attention pooling*
- **Positional Encoding:** self-attention has no inherent notion of token order → absolute or relative position information must be explicitly injected

Basic Structure of the Encoder

- Several **identical layers** arranged sequentially
- Each layer: **multi-head self-attention** + **position-wise feed-forward network**
- **Residual connections** around both sublayers, followed by **layer normalization**
- Output: each token → a vector representation

How Attention Weights are Computed

Compatibility (similarity score)

$$a(q, k_i)$$

Measures how relevant key k_i is for query q .

Normalization via Softmax

$$\alpha_i = \text{softmax}(a(q, k_i)) = \frac{\exp(a(q, k_i))}{\sum_{j=1}^m \exp(a(q, k_j))}$$

α_i converts raw similarity scores into probabilities + ensures $\sum_i \alpha_i = 1$ + highlights the most relevant elements

Note: $a(q, k_i)$ is a placeholder here - Transformers use a specific scaled dot-product version of it ($q^\top k_i / \sqrt{d}$), since the dot product is a simple and efficient way to measure vector similarity. The softmax + weighted sum framework above stays the same.

- if you are interested why you can have a look at the source I assigned for today and read chapter "11.3. Attention Scoring

Attention as Weighted Aggregation

Attention computes a weighted combination of value vectors based on similarity between a query and keys.

Attention Formula:

$$\text{Attention}(q, \mathcal{D}) = \sum_{i=1}^m \alpha_i v_i$$

- $\mathcal{D} = \{(k_1, v_1), \dots, (k_m, v_m)\}$: database of key-value pairs
- q : query (what we are looking for)
- k_i : keys (descriptions of items)
- v_i : values (information content)
- α_i : how relevant each item is to the query

→ mixture of value vectors, weighted by relevance to the query

Attention must work efficiently with variable-length sequences and large minibatches → we need extra functions

Masked Softmax (handling padding)

- Real sequences have different lengths (e.g., NLP sentences)
- Shorter sequences are padded with `<blank>` tokens
- These should NOT affect attention
- Solution: mask invalid positions before softmax

→ Attention weights for padding tokens become 0 + only valid tokens contribute to the output

From Static Attention to Dynamic Context (Seq2Seq View)

The issue with Seq2Seq without attention:

- Encoder compresses the entire input sequence into a single fixed vector
- Decoder relies only on this fixed representation
- Leads to information bottleneck for long sequences

Attention idea:

- Instead of one fixed context, compute a **context per decoding step**
- Use attention over all encoder hidden states
- Dynamically select relevant parts of the input at each output step

Multi-Head Attention (Extending Single Attention)

Instead of computing one attention, we compute **multiple attentions in parallel**.

- Linearly project queries, keys, and values into different subspaces
- Apply attention independently in each subspace (each is a **head**)
- Concatenate outputs from all heads
- Apply a final linear transformation

→ Different heads can focus on different patterns + richer representations than single attention

Self-Attention and Positional Encoding

Self-Attention:

Each token in a sequence attends to all other tokens in the same sequence.

- Queries, keys, and values all come from the **same input sequence**
- Each token builds its own representation via weighted aggregation of all tokens
- Output has the same shape as input (sequence-to-sequence mapping)

Key advantages vs. CNNs and RNNs:

- Fully parallel computation (no recurrence)
- Shortest path length between tokens → better long-range dependency learning
- But: computational cost is quadratic in sequence length

The Transformer Architecture (Vaswani et al., 2017)

A sequence-to-sequence model built entirely on attention → no RNNs or CNNs.

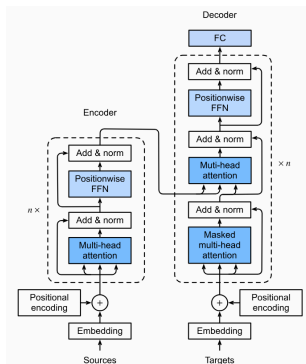


Fig. 11.7.1 The Transformer architecture.

Key Building Blocks

- **Self-attention:** dependencies between all tokens
- **Multi-head attention:** different representation subspaces
- **FFN:** per-token non-linear transformation
- **Residual + LayerNorm:** stabilizes deep training

Decoder-Specific

- **Masked self-attention:** prevents seeing future tokens
- **Encoder-decoder attention:** aligns output with input

How can we use Transformers for images?

→ *treat image patches as tokens*

Image → Sequence of Patch Embeddings

- Split image into fixed-size patches (e.g. 16×16 px, flattened & linearly projected into a vector)
- Add positional embeddings (preserve spatial structure)
- Add special [CLS] token for classification

Architecture

- **Patch embedding:** patches → token sequence
- **Transformer encoder:** global relationships via self-attention
- **MLP head:** [CLS] token → prediction

Limitations

- Self-attention is quadratic in patches → expensive for high-resolution images
- Needs large datasets to outperform CNNs

Discussion

Do you notice any remaining issues we did not have for CNNs?

Vision Transformers (ViT)

Positional Encoding:

- Self-attention has no built-in notion of order
- Positional encodings inject sequence order into token embeddings

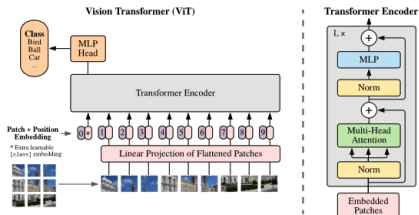


Figure: Taken from Dosovitskiy et al. 2021

Thanks for your attention!

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